

Multiple Choice (10 marks)

1. Suppose we like to calculate $P(H|E, F)$ and we have no conditional independence information. Which of the following sets of numbers are sufficient for the calculation?
 - (a) $P(E, F), P(H), P(E|H), P(F|H)$
 - (b) $P(E, F), P(H), P(E, F|H)$
 - (c) $P(H), P(E|H), P(F|H)$
 - (d) $P(E, F), P(E|H), P(F|H)$
2. Consider the Bayesian network given below. How many independent parameters would we need if we made no assumptions about independence or conditional independence $H \rightarrow U \leftarrow P \leftarrow W$?
 - (a) 3
 - (b) 4
 - (c) 7
 - (d) 15
3. The number of test examples needed to get statistically significant results should be __
 - (a) Larger if the error rate is larger.
 - (b) Larger if the error rate is smaller.
 - (c) Smaller if the error rate is smaller.
 - (d) It does not matter.
4. Which of the following best describes the joint probability distribution $P(X, Y, Z)$ for the given Bayes net. $X \leftarrow Y \rightarrow Z$?
 - (a) $P(X, Y, Z) = P(Y) * P(X|Y) * P(Z|Y)$
 - (b) $P(X, Y, Z) = P(X) * P(Y|X) * P(Z|Y)$
 - (c) $P(X, Y, Z) = P(Z) * P(X|Z) * P(Y|Z)$
 - (d) $P(X, Y, Z) = P(X) * P(Y) * P(Z)$
5. Adding more basis functions in a linear model, pick the most probably option:
 - (a) Decreases model bias
 - (b) Decreases estimation bias
 - (c) Decreases variance
 - (d) Doesn't affect bias and variance

True / False (10 marks)

Answer True or False

6. True or False: The softmax function is commonly used in multiclass logistic regression to convert logits into probabilities.
7. True or False: DenseNets typically require more memory than ResNets due to their densely connected architecture.
8. True or False: The variance of the Maximum A Posteriori (MAP) estimate is the same as that of the Maximum Likelihood Estimate (MLE).
9. True or False: Neural networks can only be effectively trained using stochastic gradient descent as their optimization method.
10. True or False: $P(A \mid B, C) * P(B \mid A, C) * P(C \mid A, B)$ is a correct formulation for $P(A, B, C)$ in the absence of independence assumptions.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the effectiveness of convolutional networks in classifying high-resolution images, highlighting their architectural features, advantages, and any limitations compared to other approaches.
12. Discuss the implications and methodologies of using unsupervised learning to identify distinct patient clusters in a dataset of medical records for heart disease treatment.

End of Paper